Higher Hida theory and the p-adic L-function for $\mathrm{U}(2,1)$

Gyujin Oh

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Outline

1. Introduction
2. Higher Hida theory for $\mathrm{U}(2,1)$
3. Integral representation of the L-function
4. Construction of the p-adic L-function

Periods and L-functions

For an automorphic representation π of G, there are various numerical values attached to them,

$$L(s, \pi, \rho) \qquad \text{special value of } L\text{-function}$$

$$\int_{[H]} f(h)dh, f \in \pi \qquad \text{period}$$

For L-values, there are very general conjectures on their transcendental part (e.g. Beilinson's conjecture), and one can hope that the algebraic part satisfies p-adic congruence property as we "p-adically vary" either s or π.

For periods, they are oftentimes connected to L-values, so by similar reason one can expect p-adic interpolation of periods.

Refined Gan–Gross–Prasad conjectures

A major source of relations between periods and L-values is the **Gan–Gross–Prasad conjecture**. Its refined form states, roughly, that, for classical groups $H \subset G$ of the same type, and for $f \in \pi$ and $f' \in \pi'$ cusp forms on G and H, respectively,

$$\left| \int_{[H]} f(h) f'(h) dh \right|^2 = C \frac{L^S(\frac{1}{2}, \pi \otimes \pi')}{L^S(1, \pi, \mathrm{Ad}) L^S(1, \pi', \mathrm{Ad})} \prod_{v \in S} I_v(f_v, f'_v),$$

for an explicit constant C and some local factors I_v.

In this regard, the GGP period integral is also regarded as a "square root of an L-function."

Refined Gan–Gross–Prasad conjectures: Degenerate case

It is also understood that the same period for $f' \in \pi'$ Eisenstein is related to L-values,

$$\int_{[H]} f(h)E(h,s)dh = C_\pi C_{\pi'} L^S(\pi, s) \prod_{v \in S} \alpha_v(f_v, s).$$

If f is generic, C_π should be related to the Whittaker period of π (Lapid–Mao conjecture).

This type of formula is established in several cases (Novodvorsky, Piatetski-Shapiro for Sp_4, Gelbart–Piatetski-Shapiro for $\mathrm{U}(2,1)$, Tsuzuki for $\mathrm{U}(n,1)$, Pollack for $\mathrm{O}(n,2)$, $\cdots$).

Main objective

Our objective is to use the integral for $\mathrm{U}(2,1)$ to construct a p-adic L-function using coherent cohomology of Shimura varieties, following the strategy of Loeffler–Pilloni–Skinner–Zerbes.

Let $G = \mathrm{U}(2,1)$ be defined using an imaginary quadratic field K on which $p = \pi\overline{\pi}$ is split.

Main Theorem

Suppose Π is a generic, stable, cuspidal automorphic representation of $\mathrm{GU}(2,1)$ of infinitesimal character (k_1, k_2, k_3), which is Klingen-ordinary and unramified at π. If $k_1 - k_2 \geq 2$ and $k_2 + k_3 \geq 2$, there is an element $\mathcal{L}(\Pi, \mathbf{j}) \in \Lambda_{\overline{\mathbb{Q}}_p}(\mathbb{Z}_p^\times)$, the Iwasawa algebra, such that, for any Dirichlet character χ of p-power conductor and any critical value a of $L(\Pi, \chi, s)$,

$$\mathcal{L}(\Pi, a + \chi) = Z_\infty(\Pi, \chi, a) Z_p(\Pi, \chi, a) L(BC(\Pi) \otimes \chi^{-1}, a),$$

where Z_∞ and Z_p are certain local factors at ∞ and p, respectively.

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Higher Hida theory

The starting point is that the Gelbart–Piatetski-Shapiro integral can be regarded as a cup product in coherent cohomology of Shimura variety for $\mathrm{U}(2,1)$.

$$L(BC(\pi), s) \sim \int_{[\mathrm{U}(1,1)]} f(h)E(h,s)dh \sim \langle f, \iota_*(E)\rangle,$$

where $\iota : X_{\mathrm{U}(1,1)} \to X_{\mathrm{U}(2,1)}$ is the natural embedding of Shimura varieties.

As $\iota_*(E) \in H^1(X_{\mathrm{U}(2,1)})$, f should be in H^1. The p-adic interpolation of higher coherent cohomology is the subject of **higher Hida theory**, which generalizes the classical Hida theory on p-adic interpolation of H^0 (holomorphic automorphic forms).

Igusa tower

Note that $Y_{\mathrm{GU}(2,1)}$ is the moduli space of abelian threefolds with PEL structure, most notably equipped with an action of K up to isogeny. The classical Hida theory involves the cohomology of the sheaf coming from the Igusa tower $\mathrm{Ig}(p^\infty)$ over the ordinary locus, namely the isomorphism

$$\mathcal{A}[\pi^\infty]^\circ \cong \mu_{p^\infty} \times \mu_{p^\infty}.$$

However, the ordinary locus is always **affine**, so its higher coherent cohomology vanishes. The ordinary locus is affine because it is the nonvanishing locus of a section (**Hasse invariant**) of an ample line bundle over a projective variety. Thus, to have nonzero H^i, one should look for removing a "depth i" locus.

Ekedahl–Oort stratification

Such structures can be given by the **Ekedahl–Oort stratification**. It is stratified by the isomorphism classes of $A[p]$ as a finite flat group scheme. Most notably, the closure of each stratum is a union of strata, and over the closure, there is a distinguished section, **generalized Hasse invariant**, of an ample line bundle, whose nonvanishing locus is the stratum we started with.

In our case, we have three strata: the ordinary locus ($A[\pi]$ has multiplicative rank 2), the multiplicative rank 1 locus $X^{=1}$, and the multiplicative rank 0 locus $X^{=0}$.It is nested as

$$\overline{X^{=i}} = X^{\leq i},$$

and $X^{=i}$ is the nonvanishing locus of a generalized Hasse invariant, a section over $X^{\leq i}$. In particular, $X^{\geq 1} = X - X^{=0}$ has nonvanishing H^0 and H^1.

Partial Igusa tower

In contrast to the Igusa tower $\mathrm{Ig}(p^\infty)$ over the ordinary locus, one can define a **partial Igusa tower** $\pi : \mathrm{Ig}^{\geq 1}(p^\infty) \to X^{\geq 1}$, which parametrizes $\mu_{p^\infty} \hookrightarrow \mathcal{A}[\pi^\infty]$. This is a tower with an action of $\mathbb{Z}_p^\times$ (in contrast to $\mathrm{Ig}(p^\infty)$ possessing an action of $\mathbb{Z}_p^\times \times \mathbb{Z}_p^\times$).

Analogously to the Hida theory, one is led to consider the cohomology of the sheaf

$$\mathfrak{F} := \pi_* \mathcal{O}_{\mathrm{Ig}^{\geq 1}(p^\infty)} \widehat{\otimes}_{\mathbb{Z}_p} \Lambda,$$

where $\Lambda = \mathbb{Z}_p[[\mathbb{Z}_p^\times]]$ is the Iwasawa algebra. The complex $R\Gamma(X^{\geq 1}, \mathfrak{F})$ is a complex of Λ-modules, which, hopefully, interpolates H^0 and H^1 of X.

In search of optimally p-integral Hecke operator

As in Hida theory, $R\Gamma(X^{\geq 1}, \mathfrak{F})$ is too big, and it needs to be cut down by taking the ordinary part.

The T_p-operator can be given by a geometric correspondence

$$\begin{tikzcd} & X(K^pP) \arrow[dl] \arrow[dr] & \\ X(K^pG(\mathbb{Z}_p)) & & X(K^pG(\mathbb{Z}_p)) \end{tikzcd}$$

for a "Klingen parahoric" P. To take the ordinary part, we need to know the minimum of valuations of T_p-eigenvalues. This requires studying the geometry of the above diagram. Knowing this, we can define the ordinary projector e_{Kl}.

Hida complex

We define the **Hida complex** to be $M^\bullet = e_{\mathrm{Kl}} R\Gamma(X^{\geq 1}(p), \mathfrak{F})$.

Control theorem

Certain automorphic vector bundles are defined as

$$\omega_G(k_1, k_2, k_3) := \mathrm{Sym}^{k_1-k_2}\,\omega_\sigma \otimes (\det \omega_\sigma)^{k_2} \otimes \omega_{\overline{\sigma}}^{k_3},$$

where, for $\tau : K \to \overline{\mathbb{Q}}$, and ω_τ is the τ-part of $\mathscr{H}^1_{\mathrm{dR}}(\mathcal{A}/Y)$.

Theorem (Integral control theorem; Nguyen)

For $k_1 > k_2$, $k_1 + k_3 > 3$,

$$M^\bullet \otimes^L_{\Lambda, k_1} \mathbb{Z}_p \cong e_{\mathrm{KI}} R\Gamma(X(K^p G(\mathbb{Z}_p))_{\mathbb{Z}_p}, \omega(k_1, k_2, k_3)).$$

Remarks

1. This is proven in several steps, by first relating the cohomology of $X^{\geq 1}(p)$ (Klingen) with that of $X^{\geq 1}$ (hyperspecial), and then relating it with that of X, by showing that T_p is divisible by p on $R\Gamma(X_1^{=0})$.
2. We have a better handle on the integral control theorem than Pilloni's GSp_4 case, because, for GSp_4, the Hecke operator is artifically realized as the composition of two geometric correspondences.

Maps between partial Igusa towers

Our objective is to p-adically interpolate the cup product pairing. To achieve this, we need to relate the Hida theory (H^0) of X_H for $H = \mathrm{GU}(1,1) \times_{\mathrm{GL}_1} \mathrm{GU}(1)$ and the higher Hida theory of X_G.

Over X_H^{ord}, we define the partial Igusa tower $\mathrm{Ig}_H^{\mathrm{ord}}(p^\infty)$ to be the pullback of the Igusa tower for $X_{\mathrm{GU}(1,1)}^{\mathrm{ord}}$.

Maps between partial Igusa towers

There is a natural tower of schemes

$$\begin{tikzcd} \mathrm{Ig}_H^{\mathrm{ord}}(p^m) \arrow[r, "\widetilde{\iota}_m"] \arrow[d] & \mathrm{Ig}_G^{\geq 1}(p^m) \arrow[d] \\ X_H^{\mathrm{ord}}(p^m) \arrow[r, "\iota_m"] & X_G^{\geq 1}(p^m) \end{tikzcd}$$

sending $((A_1, C), A_2) \mapsto (A_1 \times A_2, C)$.

p-adic pushforward map

Maps between partial Igusa towers

For any $k_1 \geq \kappa_1 \geq k_2 \geq \kappa_2 \geq -k_3$, there is a commutative diagram over $X_H^{\mathrm{ord}}(p)$,

$$\begin{tikzcd}
\omega_H(\kappa_1, \kappa_2; -4) \arrow[r, hook] \arrow[d] & \iota_1^* \omega_G(k_1, k_2, k_3) \arrow[d] \\
\mathfrak{F}_H^{(\kappa_1,\kappa_2)} \otimes \omega_H(0, 0; -4) \arrow[r, hook] & \iota_1^* \mathfrak{F}_G^{(k_1,k_2,k_3)}
\end{tikzcd}$$

Composing this with the pushforward

$$\iota_* : H^0(X_H^{\mathrm{ord}}(p), \iota_1^* \mathfrak{F}_G^{(k_1,k_2,k_3)} \otimes \omega_H(1, -2; 0)) \to H^1(X_G^{\geq 1}(p), \mathfrak{F}_G^{(k_1,k_2,k_3)}),$$

we get the p-adic pushforward from the Hida theory of X_H to the higher Hida theory of X_G,

$$\iota_* : \mathcal{S}_{(\kappa_1-1,\kappa_2+2)}(R) \to H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{\kappa}(-D_G)).$$

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Integral formula of Gelbart–Piatetski-Shapiro

Let Π be a cuspidal automorphic representation of $\mathrm{GU}(2,1)(\mathbb{A}_{\mathbb{Q}})$, and $f \in \Pi$. Let μ be an idele class character of $K^\times \backslash \mathbb{A}_K^\times$. Let Φ be a Schwartz–Bruhat function on $V_H(\mathbb{A}_K)$. Then, we would like to consider the zeta integral

$$Z(f, \mu, \Phi, s) := \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} f(h) E^\Phi(h; \mu, s) dh,$$

where

$$E^\Phi(h; \mu, s) := \sum_{\gamma \in B_H(\mathbb{Q})\backslash H(\mathbb{Q})} f^\Phi(\gamma h; \mu, s),$$

$$f^\Phi(h; \mu, s) := \int_{\mathbb{A}_K^\times} (h \cdot \Phi)(ty) \mu(t) |t|_K^s d^\times t \qquad \text{(“Mellin transform”).}$$

Unfolding of GPS integral

Suppose $\Phi = \prod_w \Phi_w$ is factorizable. Then

$$\begin{aligned} Z(f,\mu,\Phi,s) &= \int_{Z_G(\mathbb{A})Z_{N_{B_G}}(\mathbb{A})\backslash H(\mathbb{A})} W_f^{\psi}(h) f^{\Phi}(h;\mu,s)dh \\ &= \prod_v \int_{Z_G(\mathbb{Q}_v)Z_{N_{B_G}}(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} W_v(h) f_v^{\Phi}(h;\mu,s)dh \\ &=: \prod_v Z_v(f,\mu_v,\Phi_v,s), \end{aligned}$$

where

- $Z_{N_{B_G}}$ is the center of the unipotent radical of B_G,
- ψ is a nontrivial additive character of $\mathbb{Q}\backslash\mathbb{A}_{\mathbb{Q}}$,
- $W_f^{\psi}(g) = \int_{N_{B_G}(\mathbb{Q})\backslash N_{B_G}(\mathbb{A}_{\mathbb{Q}})} f(ng)\overline{\psi_{N_{B_G}}(n)}dn$,
- $W_v(h)$ is in the local Whittaker model $W(\pi_v,\psi_v) = \{W_v : G(\mathbb{Q}_v) \to \mathbb{C} \mid W_v(ng) = \psi_v(n)W_v(g) \forall n \in N_{B_G}(\mathbb{Q}_v)\}$, which is at most one-dimensional,
- $f_v^{\Phi}(h;\mu,s) = \prod_{w|v} \int_{K_w^\times} (h\cdot\Phi_w)(ty)\mu_w(t)|t|_w^s d^\times t$.

Unfolding of GPS integral

The unfolding of GPS integral uses the Fourier–Jacobi expansion,

$$f(h) = f_{00}(h) + \sum_{\psi \neq 1} f^{\psi}(h),$$

where the sum runs over nontrivial characters of $[Z_{N_{B_G}}]$, $f_{00}(h) = \int_{[Z_{N_{B_G}}]} f(nh)dn$, and

$$f^{\psi}(h) = \int_{[Z_{N_{B_G}}]} f(zh)\overline{\psi(z)}dz,$$

is the "Fourier–Jacobi coefficient." The key is that only f_{00} term survives, as after the unfolding of Eisenstein series, $Z_{N_{B_G}}$ acts trivially. Then f_{00} can be expressed in terms of Whittaker model of f.

Generalized Whittaker models

If the Eisenstein series was replaced by a cusp form g, then this means the integral of g over $[Z_{N_{B_G}}]$ is zero, so the contribution rather comes from $f^{\psi}(h)$. The problem is that $f^{\psi}(h)$ is not supported by a multiplicity one model. Indeed, it is supported by

$$\mathrm{Ind}_{N_{B_G}}^{G} \omega_{\psi},$$

where ω_{ψ} is the Weil representation with central character ψ. It is infinite-dimensional as we are integrating over a very small unipotent group $Z_{N_{B_G}}$, which is smaller than the unipotent group N_{B_G} (**wavefront**) which supports a nonzero model (Whittaker model, as it is generic). This is called a **degenerate Whittaker model**.

It is a rather surprising fact that infinite-dimensional degenerate Whittaker models can be salvaged by considering the action of an extension of unipotent group by a reductive group. This is called a **generalized Whittaker model**. The first systematic development is due to Yamashita.

Unfolding of GGP period integral, $\mathrm{U}(2,1) \times \mathrm{U}(1,1)$

In the case of $\mathrm{U}(2,1)$, we need to consider the transformation property under the action of

$$\mathrm{U}(1) \ltimes N_{B_G},$$

whose irreducible unitary representations are of form $\chi \otimes \omega_\psi$.

Theorem (Multiplictiy one property of generalized Whittaker models; Gelbart–Rogawski, Ishikawa)

The representation $\mathrm{Ind}_{\mathrm{U}(1)\ltimes N_{B_G}}^{G}(\chi \otimes \omega_\psi)$ is multiplicity-free.

Now it is not hard to unfold the GGP period integral for (f, g) to the generalized Whittaker model of f and the Whittaker model of g

$$\begin{aligned} Z(f,g,\mu) &:= \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} f(h)g(h)\mu(h)dh \\ &= \int_{Z_H(\mathbb{A})N_{B_H}(\mathbb{A})\backslash H(\mathbb{A})} f^{\psi,\mu^{-1}}(h)W_g^{\psi}(h)dh. \end{aligned}$$

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Nearly holomorphic modular forms

It is a general wisdom that one needs the period of **nearly holomorphic modular forms** to access all critical values of L-functions. Nearly holomorphic modular forms are obtained from holomorphic modular forms via applying Maass–Shimura raising operators.

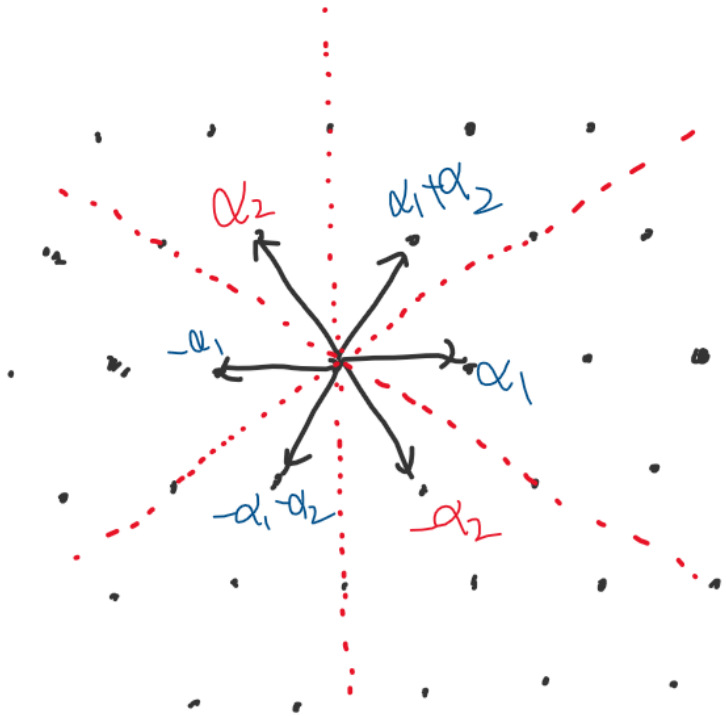
It has a very simple interpretation in the context of $(\mathfrak{g}, K)$-modules; it is a vector in a non-minimal K-type. It is obtained from the minimal K-type by applying the action of an element in $U(\mathfrak{p}^+)$ (holomorphic tangent space). This can be approached algebro-geometrically via using more general automorphic vector bundles.

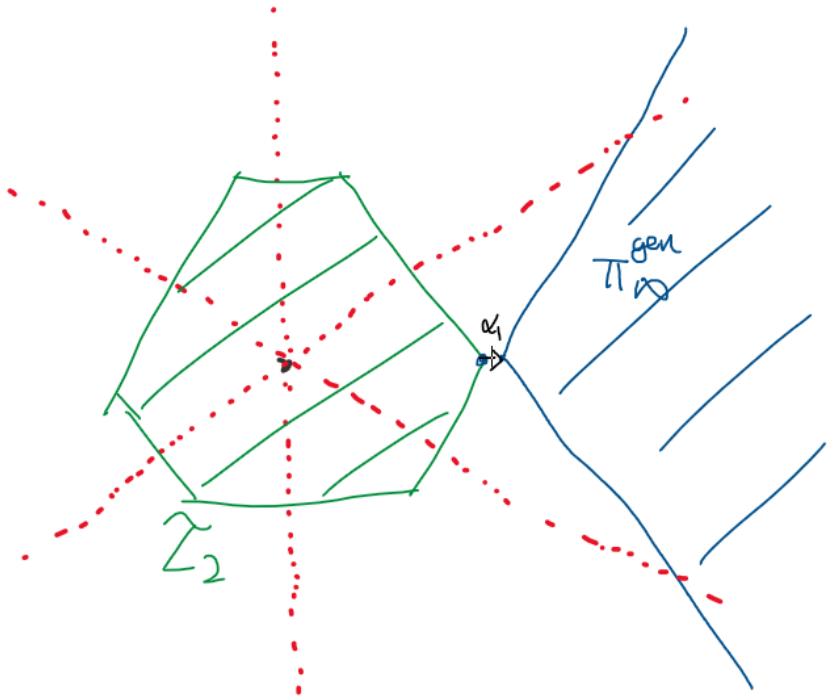
Automorphic cohomology

Automorphic vector bundles on a Shimura variety originate from vector bundles on the corresponding period domain. Using a more suggestive notation, it comes from the P_--torsor on the flag variety G/P_- via the Borel embedding $G(\mathbb{R})/K \hookrightarrow G/P_-$. In general, one can take any algebraic P_--representation and make it into a vector bundle over G/P_-. We call these **nearly sheaves**.

This process yields the same vector bundles as those coming from $K_{\mathbb{C}}$-representations, but encodes more refined information than just the isomorphism classes as coherent sheaves, e.g. Hecke action, $U(\mathfrak{g})$-action. On the other hand, one needs to make sure that there is no unwanted extra weight by checking there is no extra overlap with the K-types of the automorphic representation introduced by this.

Picture


α_2
$\alpha_1+\alpha_2$
$-\alpha_1$
α_1
$-\alpha_1-\alpha_2$
$-\alpha_2$


α_1
π_∞^{gen}
Σ_2

Archimedean splitting

Specifically, we let

$$\widetilde{L} := \left(\mathrm{Fil}^{\frac{3}{2}(k_2-k_3)}\, V_G(k_1,k_2,k_3;-k_1-k_2+k_3)\right) \otimes W(1,-1,0;0),$$

where for an algebraic representation V, Fil^n is the sum of weight spaces of weight $\lambda(k_1,k_2,k_3;c)$ with $k_1+k_2-k_3-c \geq 2n$. Let $\widetilde{L}'$ be the top filtration quotient. There is a natural map of algebraic representations of $P_-(\mathbb{C})$,

$$\widetilde{L} \twoheadrightarrow \widetilde{L}' \hookleftarrow \omega(k_1,k_2,k_3;k_1+k_2-k_3-4).$$

This is split as C^∞ K_∞-representations as K_∞ is compact, which is called the **Hodge splitting**. This gives a smooth automorphic form $H^1(X_G,\omega(k_1,k_2,k_3)_{C^\infty}) \subset H^1(X_G,[\widetilde{L}]_{C^\infty})[\Pi_f]$.

Similarly, one can define the Hodge splitting $[V_H(\kappa_1,\kappa_2)] \otimes \omega_H(1,-1) \to \omega_H(\kappa_1+1,\kappa_2-1)$ and correspondingly **nearly holomorphic** $E_\infty \in H^0(X_{H,\mathbb{C}},\omega(\kappa_1,\kappa_2-1)_{C^\infty})$ from an Eisenstein series $E \in H^0(X_{H,\mathbb{C}},[V_H]\otimes\omega_H(1,-1))$.

Archimedean splitting

Given $\eta \in H^1(X_{G,\overline{\mathbb{Q}}}, \omega(k_1, k_2, k_3))[\Pi_f]$, let $\widetilde{\eta} \in H^1(X_{G,\overline{\mathbb{Q}}}, [\widetilde{L}_1](-D))[\Pi_f]$ be the corresponding class under the above isomorphism

$$H^1(\omega(k_1, k_2, k_3))[\Pi_f] \to H^1([\widetilde{L}_1'](-D))[\Pi_f] \leftarrow H^1([\widetilde{L}_1](-D))[\Pi_f].$$

Proposition

Let $F_\eta : \tau_1 \to \Pi$ be the K-type corresponding to η. Then,

$$\langle \iota_*^{\mathrm{nearly}}(E), \widetilde{\eta} \rangle = \frac{1}{(2\pi i)^2} \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} F_\eta(v_{\kappa_1,\kappa_2}) E_\infty(h) dh,$$

where

$$\iota_*^{\mathrm{nearly}} : H^0(X_{H,\mathbb{Q}}, [V_H] \otimes \omega_H(1, -1, 0; 0)) \to H^1(X_{G,\mathbb{Q}}, [\widetilde{L^\vee}_1])$$

is the pushforward of nearly sheaves.

p-adic splitting

We are thus left with p-adically interpolating $\langle \iota_*^{\text{nearly}}(E), \widetilde{\eta} \rangle$ by putting a suitable Eisenstein series E into a family of Eisenstein series $\mathcal{E}$. Since $H = \mathrm{GU}(1,1)$, the canonical model of H-Shimura variety is a (union of) modular curves, so one can more or less use the formula of Katz's real analytic Eisenstein series keeping the same rationality property.

One still needs to show that there is a suitable p-adic interpolation of period pairing which interpolates the automorphic period pairing. The analogue of Hodge splitting in the p-adic case is the unit root splitting,

$$0 \to \omega_A \to \mathcal{H}^1_{\mathrm{dR}}(A) \to \omega_A^\vee \to 0 \quad \rightsquigarrow \quad \mathcal{H}^1_{\mathrm{dR}}(A) \cong \omega_A \oplus \mathcal{U}.$$

This happens over the ordinary locus X_G^{ord}, but Iovita's recast of the unit root splitting enables us to come up with the "unit root splitting" over $X_G^{\geq 1}$.

p-adic splitting

More specifically, Iovita's unit root splitting gives us the following. From $i : \mu_{p^\infty} \hookrightarrow A[\pi^\infty]$ over $\mathrm{Ig}^{\geq 1}(p^\infty)$, we get $\omega_\sigma \xrightarrow{i^*} \mathcal{O}_{\mathrm{Ig}(p^\infty)}$, or $[i^*] \in H^0(\mathrm{Ig}^{\geq 1}(p^\infty), \omega_\sigma^\vee)$. The unit root splitting gives a canonical lift $[i^*] \in H^0(\mathrm{Ig}^{\geq 1}(p^\infty), \mathcal{H}^1(A)_{\mathrm{dR},\overline{\sigma}})$.

p-adic splitting

Let $\mathcal{T}^{\mathrm{KI}}$ be the $P_- \cap \overline{B} = \begin{pmatrix} * & & \\ * & * & \\ & & * \end{pmatrix}$-torsor over $\mathfrak{X}_{\overline{G}}^{\geq 1}(p^\infty)$ parametrizing bases f_1, f_2, f_3 of $\mathcal{H}^1_{\mathrm{dR}}(A)_{\overline{\sigma}}$ compatible with the filtration, with the additional property that the pullback of f_3 to $\mathfrak{IG}_{\overline{G}}^{\geq 1}(p^\infty)$ is a scalar multiple of $[i^*]$.

This torsor is a reduction of the P_--torsor we used to define automorphic vector bundles. This enables us to define p-adic splitting of representations.